Bootstrap Utilities and Helpers (Bootstrap 5)

In **Bootstrap 5**, Utilities are low-level, single-purpose CSS classes that help you quickly apply styling without writing custom CSS.
Helpers are small utility-like classes designed for specific UI behaviors (clearfix, position helpers, etc.).

They follow a **utility-first approach** similar to Tailwind, but inside the Bootstrap ecosystem.

1️⃣ Spacing Utilities (Margin & Padding)

Bootstrap uses this formula:

{property}{sides}-{breakpoint}-{size}

Property

- m → margin
- p → padding

Sides

- t → top
- b → bottom
- s → start (left)
- e → end (right)
- x → left & right
- y → top & bottom
- blank → all sides

Sizes

- 0 to 5
- auto

Example:

```
<div class="mt-3 p-4 bg-light">
  Spacing Example
```

```
</div>
```

- mt-3 → margin-top
- p-4 → padding all sides

2️⃣ Color Utilities

Bootstrap provides contextual color classes.

```
<p class="text-primary">Primary text</p>

<p class="text-danger">Danger text</p>


<div class="bg-warning text-dark p-3">
  Background warning
</div>
```

Common colors:

- primary
- secondary
- success
- danger
- warning
- info
- light
- dark

3️⃣ Display Utilities

Control element visibility and display.

```
<div class="d-none">Hidden</div>

<div class="d-block">Block</div>
```

```
<div class="d-flex">Flex container</div>
```

Responsive example:

```
<div class="d-none d-md-block">
  Visible only on md and above
</div>
```

4️⃣ Flex Utilities

Bootstrap gives powerful flex helpers.

```
<div class="d-flex justify-content-between align-items-center">
  <div>Left</div>
  <div>Right</div>
</div>
```

Common classes:

- justify-content-start
- justify-content-center
- justify-content-end
- align-items-start
- align-items-center
- flex-column
- flex-row

5️⃣ Text Utilities

```
<p class="text-center">Centered</p>
<p class="text-uppercase">UPPERCASE</p>
<p class="fw-bold">Bold text</p>
<p class="fst-italic">Italic</p>
```

6️⃣ Border Utilities

```
<div class="border border-primary p-3">
  Border example
</div>


<div class="rounded p-3 border">
  Rounded border
</div>
```

Options:

- border-top
- border-0
- rounded-circle
- rounded-pill

7️⃣ Position Utilities

```
<div class="position-relative">
  <div class="position-absolute top-0 end-0">
    Top Right
  </div>
</div>
```

Classes:

- position-static
- position-relative
- position-absolute
- position-fixed

- top-0
- bottom-0
- start-0
- end-0

8️⃣ Shadow Utilities

```
<div class="shadow p-4 mb-3 bg-white">
  Default shadow
</div>

<div class="shadow-lg p-4 bg-white">
  Large shadow
</div>
```

9️⃣ Width & Height Utilities

```
<div class="w-50 bg-info">50% width</div>
<div class="h-100 bg-warning">100% height</div>
```

🔟 Overflow Utilities

```
<div class="overflow-auto" style="height:100px;">
  Long content here...
</div>
```

Options:

- overflow-hidden
- overflow-scroll
- overflow-auto

1️⃣1️⃣ Visibility Helpers

```
<div class="visible">Visible</div>

<div class="invisible">Invisible but space reserved</div>
```

1️⃣2️⃣ Clearfix Helper

Used when floating elements:

```
<div class="clearfix">
  <div class="float-start">Left</div>
  <div class="float-end">Right</div>
</div>
```

🔥 Why Utilities Are Powerful

✔ Faster development
✔ Less custom CSS
✔ Responsive ready
✔ Clean & consistent design
✔ Easy maintenance